

Assignment 6: Simulating the Ising Model

26th January, 2026

IMPORTANT INSTRUCTIONS

- This is a coding project. You will write Python code to simulate and visualize.
- **SUBMISSION FORMAT:**
 - Create one Notebook (.ipynb) file
 - Include markdown cells with section titles
 - Add comments in code to explain your logic
 - Upload to Google Classroom

For queries contact us on Whatsapp.

PROJECT OVERVIEW

In this project, you will build a simple simulation of how spins behave on a grid at different temperatures. Think of it as a game where:

You have a checkerboard with squares. Each square contains a tiny arrow pointing either up or down. When temperature is low, nearby arrows want to point the same way (to save energy). When temperature is high, arrows flip randomly and ignore their neighbors. Your task is to simulate this, watch patterns emerge, and create pictures showing what happens.

Even if you do not understand the physics deeply, just follow the algorithm step by step.

WHAT YOU WILL BUILD

Part 1: The Code Structure

Your notebook should have these clear sections:

1. Import libraries (numpy, matplotlib)
2. Create a grid of random spins (size 50x50 is good)
3. Write a function to calculate energy difference when one spin flips
4. Write the Metropolis algorithm (decide if flip is accepted)
5. Run simulation at multiple temperatures and save final patterns
6. Create visualizations

Part 2: Run at Different Temperatures

Run your simulation at exactly these 5 temperatures:

- Temperature = 0.5 (very cold, spins organize nicely)
- Temperature = 1.0 (cold, some flipping)
- Temperature = 2.0 (medium)
- Temperature = 2.5 (warm, more random)
- Temperature = 4.0 (hot, complete chaos)

For each temperature, run at least 5000 steps so the pattern becomes clear.

Part 3: Create 2 Visualizations

1. **Grid Comparison:** Show 5 images in a row. Each image is the final spin grid at one temperature. Color: white for spin up, black for spin down. Label each temperature. This shows how the pattern changes from ordered (cold) to random (hot).
2. **One Temperature Over Time:** Pick cold temperature (0.5 or 1.0). Show 4 snapshots: after 500 steps, 1500 steps, 3000 steps, 5000 steps. Arrange in a 2x2 grid. This shows how the pattern develops from random start to final state.
3. **Describe in a Markdown Cell:** : What do you see at cold vs hot? Where does the change happen?

WHAT RESULTS SHOULD LOOK LIKE

At cold temperature ($T = 0.5$ or 1.0): You see large white or black regions. The spins form clusters and stay organized. It looks clean and structured.

At medium temperature ($T = 2.0$ or 2.5): You see mixed regions. Some organization but also randomness. Patches of white and black scattered around.

At hot temperature ($T = 4.0$): The grid looks like television static or noise. White and black pixels scattered randomly everywhere. No obvious pattern.

This change from organized to random is the phase transition. It happens somewhere between temperature 1.0 and 2.5.

Hint [You don't need to follow this section rigidly. You are free to implement in your own way.]

Here is exactly what your code should do, line by line:

Step 1: Start with a random grid

Create a 50x50 array where each cell is randomly either +1 or -1.

Step 2: Calculate energy change if we flip one spin

If you flip a spin at position (i,j) , the new energy is based on its neighbors (up, down, left, right). Use this formula:

$$dE = 2 * J * \text{spin}[i,j] * (\text{spin}[i-1,j] + \text{spin}[i+1,j] + \text{spin}[i,j-1] + \text{spin}[i,j+1])$$

where $J = 1$ (just use 1, no need to change it). The -1 indices wrap around (use `modulo` or `np.roll()`).

Step 3: Decide if we accept the flip

Generate random number r between 0 and 1.

If dE is less than 0 (flip reduces energy): always accept the flip.

If dE greater than or equal to 0 (flip increases energy): accept only if r is less than $\exp(-dE / T)$, where T is temperature.

If accept: flip the spin. If reject: leave it unchanged.

Step 4: Repeat many times

Do steps 2-3 about 50 times per temperature run (roughly visiting each spin once per step).

Step 5: Visualize

Use `plt.imshow()` to show the grid. Use `cmap='binary'` to make +1 white and -1 black.

SUBMISSION CHECKLIST

Before submitting, check:

- Code runs without errors
- Grid comparison showing 5 temperatures is clear and labeled
- Time evolution (4 snapshots) shows pattern change clearly
- You have written comments explaining your code
- You wrote 3-4 sentences describing: What do you see at cold vs hot? Where does the change happen?